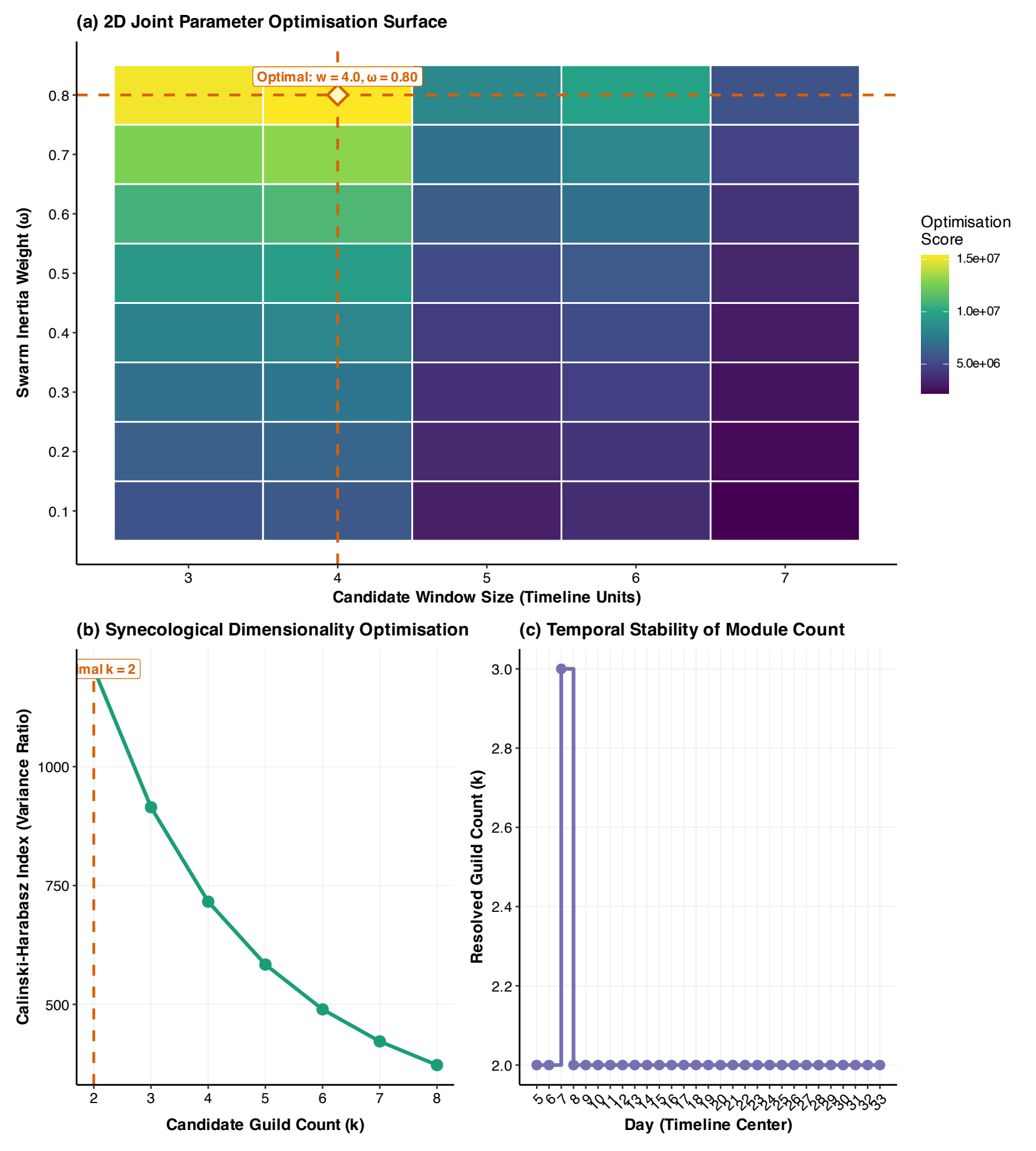
(a) 2D Joint Parameter Optimisation Surface
Optimal: w = 4.0, ω = 0.80
Swarm Inertia Weight (ω)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
3
4
5
6
7
Candidate Window Size (Timeline Units)
Optimisation Score
1.5e+07
1.0e+07
5.0e+06
(b) Synecological Dimensionality Optimisation
imal k = 2
Calinski-Harabasz Index (Variance Ratio)
1000
750
500
2
3
4
5
6
7
8
Candidate Guild Count (k)
(c) Temporal Stability of Module Count
Resolved Guild Count (k)
3.0
2.8
2.6
2.4
2.2
2.0
Day (Timeline Center)